

5.6 Estimating the error variance

The covariance formula $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is of little use until σ^2 is estimated. Least squares does not estimate it: the criterion $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ involves only $\boldsymbol{\beta}$. But the size of the residuals clearly carries information about the size of the errors. This section finds the right way to turn one into an estimate of the other, and shows why the obvious way is wrong.

5.6.1 The residual sum of squares

Since $\sigma^2 = \text{Var}(\varepsilon_i) = E(\varepsilon_i^2)$, a natural first idea is to replace each unobserved error by its residual and average: SSE/n . This underestimates σ^2 . The fitted values were chosen to make the residuals as small as possible, so the residuals are systematically smaller than the errors. The next theorem measures exactly how much smaller.

Theorem 5.6.1. Unbiased estimation of the error variance

In the linear model Definition 5.3.1 with $\text{rank}(\mathbf{X}) = p < n$,

$$E(\text{SSE}) = \sigma^2(n - p).$$

Hence

$$s^2 = \frac{\text{SSE}}{n - p} = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n - p} \quad (5.6.1)$$

is an unbiased estimator of σ^2 .

Proof. By Proposition 5.4.2(e) the residual vector is $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$, whatever $\boldsymbol{\beta}$ is. Since $\mathbf{I} - \mathbf{H}$ is symmetric and idempotent, $\text{SSE} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$, a quadratic form in the error vector, which has mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{I}_n$. By Corollary 2.3.2(b),

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) + \mathbf{0}^\top (\mathbf{I} - \mathbf{H})\mathbf{0} = \sigma^2(n - p),$$

using $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$ from Proposition 5.4.2(a). ■

The computation is the case $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$ of Example (Residual sum of squares under a wrong mean) in Chapter 2, specialized to full rank.

The same result follows from Proposition 5.5.3 without quadratic forms. Each residual has mean zero and variance $\sigma^2(1 - h_{ii})$, so $E(\text{SSE}) = \sum_i E(\hat{\varepsilon}_i^2) = \sigma^2 \sum_i (1 - h_{ii}) = \sigma^2(n - p)$. This version shows *where* the shortfall comes from: each residual is less variable than its error by the factor $1 - h_{ii}$, and the leverages add up to p .

The divisor $n - p$ is the number of **residual degrees of freedom**. The name reflects the constraints $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ of Proposition 5.4.2(b): the n residuals satisfy p independent linear equations, so only $n - p$ of them can vary freely. Chapter 6 makes this precise: the residual vector lies in a subspace of dimension $n - p$, and the degrees of freedom of a sum of squares is always the dimension of the space it lives in. For the straight line $p = 2$, so $s^2 = \text{SSE}/(n - 2)$. For the

sample variance of a single sample, the model matrix is $\mathbf{1}$, $p = 1$, and Eq. 5.6.1 is the familiar divisor $n - 1$.

Corollary 5.6.2. Estimated covariance and standard errors

Under the conditions of Theorem 5.6.1, $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is an unbiased estimator of $\text{Cov}(\hat{\boldsymbol{\beta}})$, and $s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$ is an unbiased estimator of $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$. The **standard error** of $\hat{\beta}_j$ is

$$\text{se}(\hat{\beta}_j) = s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}.$$

Proof. $(\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{a}$ are constants, so the expectations are $E(s^2)(\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ and similarly for the scalar. ■

The standard error is not unbiased for the standard deviation of $\hat{\beta}_j$. By Jensen's inequality $E(s) \leq \sqrt{E(s^2)} = \sigma$, with strict inequality unless s is constant. The bias is small when $n - p$ is large, and it matters little, since the standard error is used mainly through the ratio $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$, whose exact distribution under normality is the t distribution with $n - p$ degrees of freedom (Chapter 7, Theorem 7.5.1).

Example. Standard errors for money demand

For the money demand model of Example (Money demand in Denmark), $n = 55$ and $p = 4$, so

$$s^2 = \frac{0.09253}{51} = 0.001814, \quad s = 0.04259.$$

Since the response is a logarithm, s says that money holdings typically deviate from the fitted relation by about 4 percent. The standard errors of the four coefficients are 0.5811, 0.0940, 0.3282 and 0.6911. The income coefficient 1.2958 is almost fourteen standard errors from zero and the bond rate coefficient -2.6163 about eight, but the deposit rate coefficient 0.6186 is smaller than its standard error. Whether the deposit rate belongs in the model at all is a testing question for Chapter 11.

These standard errors rest on (L2) and (L3). For a quarterly economic series (L3) is doubtful. The correlation between consecutive residuals is 0.62, which suggests that the errors are positively correlated in time, and then the formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ usually understates the true variability. Chapter 21 shows how to detect and allow for this.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)      # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid
```

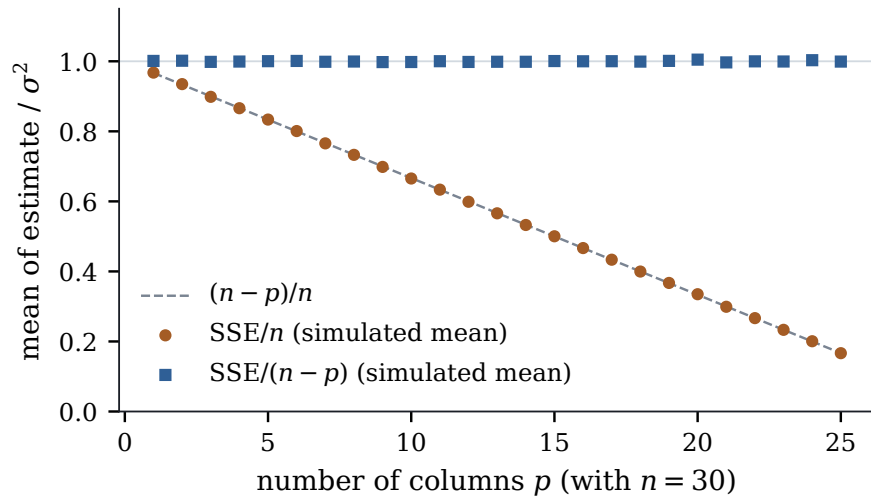


Figure 5.6.1: Average of two estimators of $\sigma^2 = 1$ over 20,000 simulated data sets, with $n = 30$ and p columns. Dividing the residual sum of squares by n underestimates σ^2 by the factor $(n-p)/n$; dividing by $n-p$ is unbiased for every p .

```
s2 = SSE / (n - p)                                # unbiased estimate of sigma^2
cov_hat = s2 * np.linalg.inv(XtX)                 # estimated Cov(beta_hat)
se = np.sqrt(np.diag(cov_hat))
print(f"s^2 = {s2:.6f}, s = {np.sqrt(s2):.5f}")
print("standard errors:", np.round(se, 4))
print("SSE / n would give", SSE / n)
```

[Run this code online](#)

5.6.2 Dividing by n

The estimator $\hat{\sigma}^2 = \text{SSE}/n$ is not unbiased, but it is not arbitrary either: under normality it is the maximum likelihood estimator (Theorem 7.3.1). Its expectation is

$$E(\hat{\sigma}^2) = \frac{n-p}{n} \sigma^2 = \sigma^2 - \frac{p}{n} \sigma^2.$$

The bias is negligible when p is a small fraction of n , and severe when it is not. For the money demand data the two estimates of σ are 0.04259 and 0.04102, a difference of about 4 percent. But a model with many columns relative to the number of cases, such as a two-way layout with a few replicates per cell or a regression with many indicator variables, can make SSE/n badly misleading.

Example. How bad is dividing by n ?

Take $n = 30$ cases and model matrices with an intercept and $p - 1$ further columns of standard normal numbers, for $p = 1, \dots, 25$. For each p , 20,000 error vectors are simulated with $\sigma^2 = 1$. Figure 5.6.1 plots the average of SSE/n and of $\text{SSE}/(n-p)$. The first follows the line $(n-p)/n$: at $p = 5$ it averages 0.834, at $p = 15$ it averages 0.500, half the true value, and at $p = 25$ only 0.167. The unbiased estimator averages 1.000, 1.001 and 0.999 at the same values of p .

```

import numpy as np
rng = np.random.default_rng(506)
n, sigma2, reps = 30, 1.0, 20_000
Xfull = np.column_stack([np.ones(n), rng.normal(size=(n, 24))])

ps = np.arange(1, 26)
mean_n, mean_np = [], []
for p in ps:
    X = Xfull[:, :p]
    R = np.eye(n) - X @ np.linalg.solve(X.T @ X, X.T)      # I - H
    E = rng.standard_normal((reps, n))                      # errors, sigma^2 = 1
    SSE = np.einsum("ri,ij,rj->r", E, R, E)                 # e^T (I - H) e
    mean_n.append(SSE.mean() / n)
    mean_np.append(SSE.mean() / (n - p))
print("p      E(SSE/n)  E(SSE/(n-p))")
for p in (1, 5, 15, 25):
    print(f"{p:2d}      {mean_n[p - 1]:.3f}      {mean_np[p - 1]:.3f}")

```

[Run this code online](#)

5.6.3 The variance of s^2

Unbiasedness says where s^2 is centred, not how much it varies. Its variance depends on the fourth moment of the errors, so the second-moment assumptions are not enough to find it. With independent errors, [Chapter 2](#) supplies the formula.

Proposition 5.6.3. Variance of s^2

In the linear model with $\text{rank}(\mathbf{X}) = p < n$, suppose the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent with mean 0, variance σ^2 and common fourth moment $\mu_4 = E(\varepsilon_i^4) < \infty$. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - p) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

For normal errors, $\mu_4 = 3\sigma^4$ and $\text{Var}(s^2) = 2\sigma^4/(n - p)$.

Proof. Apply Theorem 2.3.3 to the vector $\boldsymbol{\varepsilon}$, whose means θ_i are all zero, with $\mathbf{A} = \mathbf{I} - \mathbf{H}$. The terms involving $\boldsymbol{\theta}$ vanish (in particular the third moment enters only multiplied by $\boldsymbol{\theta} = \mathbf{0}$, so it need not even be assumed finite). The diagonal of $\mathbf{A}$ has entries $1 - h_{ii}$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - p$ because $\mathbf{A}$ is idempotent. For the normal case, the fourth moment of a $N(0, \sigma^2)$ variable is $3\sigma^4$, and $\text{Var}(s^2) = \text{Var}(\text{SSE})/(n - p)^2$. ■

The kurtosis term can be large. Heavy-tailed errors, with μ_4 much larger than $3\sigma^4$, make s^2 much less reliable than the normal-theory formula suggests, even though it remains unbiased. Since $0 \leq 1 - h_{ii} \leq 1$, the sum in the kurtosis term is at most $\sum_i (1 - h_{ii}) = n - p$, which gives the following.

Corollary 5.6.4. Consistency of s^2

Under the conditions of [Proposition 5.6.3](#),

$$\text{Var}(s^2) \leq \frac{2\sigma^4 + |\mu_4 - 3\sigma^4|}{n - p}.$$

So if $n - p \rightarrow \infty$ along a sequence of models, $s^2 \rightarrow \sigma^2$ in mean square and in probability.

Proof. Divide the formula of [Proposition 5.6.3](#) by $(n - p)^2$ and bound the sum by $n - p$. Since $E(s^2) = \sigma^2$, $E(s^2 - \sigma^2)^2 = \text{Var}(s^2) \rightarrow 0$, and Chebyshev's inequality gives convergence in probability. ■

Example. Skewed errors inflate the variance of s^2

With $n = 30$ and the first $p = 6$ columns of the model matrix of [Example \(How bad is dividing by \$n\$?\)](#), the leverages give $\sum_i (1 - h_{ii})^2 = 19.431$. For normal errors with $\sigma^2 = 1$, $\text{Var}(\text{SSE}) = 2(n - p) = 48.00$. For centred exponential errors, which have $\sigma^2 = 1$ and $\mu_4 = 9$, [Proposition 5.6.3](#) gives $48 + 6 \times 19.431 = 164.59$, and a simulation with 200,000 data sets gives 165.19. The standard deviation of s^2 is 0.289 under normal errors and 0.535 under exponential errors, nearly twice as large, although s^2 is unbiased in both cases.

```
p = 6
X = Xfull[:, :p]
H = X @ np.linalg.solve(X.T @ X, X.T)
R = np.eye(n) - H
mu4 = 9.0                                     # fourth moment of a centred Exp(1)
variable
var_normal = 2 * (n - p)
var_exp = (mu4 - 3) * np.sum(np.diag(R) ** 2) + 2 * (n - p)
reps2 = 200_000
E = rng.exponential(size=(reps2, n)) - 1.0
SSE = np.einsum("ri,ij,rj->r", E, R, E)
print(f"Var(SSE): normal errors {var_normal:.2f}, exponential errors {var_exp:.2f},"
      f" simulated {SSE.var():.2f}")
```

[Run this code online](#)

Under normality much more is true: SSE/σ^2 has the chi-squared distribution with $n - p$ degrees of freedom and is independent of $\hat{\beta}$ (Theorem 7.5.1, using Theorem 4.3.2 and Theorem 4.4.1). That is what turns $\hat{\beta}$, s^2 and the standard errors into exact confidence intervals and tests. Among unbiased estimators of σ^2 , s^2 then has the smallest variance (Theorem 7.4.6).

5.6.4 When the mean is wrong

[Theorem 5.6.1](#) assumed that $E(\mathbf{Y}) = \mathbf{X}\beta$. If instead $E(\mathbf{Y}) = \boldsymbol{\theta}$ is not of this form, [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#) shows that

$$E(\text{SSE}) = \sigma^2(n - p) + \|(\mathbf{I} - \mathbf{H})\boldsymbol{\theta}\|^2.$$

The extra term is the squared distance from the true mean vector to the nearest mean vector the model can produce. So an incomplete model *overestimates* σ^2 : variation that the model fails

to capture is counted as error. This is one reason to prefer a model that includes all relevant regressors, and it is the basis of the lack-of-fit tests of [Chapter 14](#) (Theorem 14.6.2), which compare s^2 with an estimate of σ^2 that does not depend on the form of the mean.

5.6.5 Exercises

5.6.5.1 A. Check your understanding

Exercise. A1

For regression through the origin, $E(Y_i) = \beta x_i$, what is the unbiased estimator of σ^2 ? Verify $E(\text{SSE}) = \sigma^2(n-1)$ directly, by writing $\text{SSE} = \sum_i \varepsilon_i^2 - (\sum_i x_i \varepsilon_i)^2 / \sum_i x_i^2$.

Exercise. A2

For Example (Forecasting December sea temperature), compute s and the standard error of the slope. Is the slope more than two standard errors away from 1, the value that would mean December simply tracks August one for one?

5.6.5.2 B. Practice

Exercise. B1

Show that $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$, and that for a model with an intercept $\text{SSE} = S_{yy} - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$ in the notation of Proposition 5.4.3. Explain why the second form is numerically preferable when the responses are large and nearly equal, as in Example (A curved trend and a seasonal cycle).

Solution. By Proposition 5.4.2(c), $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$, and $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ by the normal equations. For the centred form, the model with columns $[\mathbf{1}, \mathbf{X}_c]$ has the same fitted values (the proof of Proposition 5.4.3), with coefficients $(\bar{y}, \hat{\boldsymbol{\beta}}_1)$. So $\text{SSE} = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$, because the cross-product matrix is block diagonal, and $\mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = S_{yy}$. When the responses are about 340 with variation of a few units, $\mathbf{y}^\top \mathbf{y}$ and $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ are both near $n \cdot 340^2$ and agree in their leading digits; their difference loses those digits to cancellation. The centred form subtracts numbers of the size of the variation itself.

Exercise. B2

Under normality SSE/σ^2 has mean $n-p$ and variance $2(n-p)$. Among estimators $c \cdot \text{SSE}$, find the c that minimizes the mean squared error $E(c\text{SSE} - \sigma^2)^2$. Compare with $1/(n-p)$ and $1/n$.

Solution. Write $\text{SSE} = \sigma^2 Q$ with $E Q = m = n-p$ and $E Q^2 = 2m + m^2$. Then $E(cQ - 1)^2 \sigma^4 = \sigma^4(c^2(m^2 + 2m) - 2cm + 1)$, minimized at $c = m/(m^2 + 2m) = 1/(n-p+2)$. The minimum-MSE estimator divides by $n-p+2$: it is biased downward but less variable. The ordering of divisors is $n-p < n-p+2$ and, when $p \leq 2$, $n \leq n-p+2$, so for small p the maximum

likelihood divisor n lies between the unbiased and the minimum-MSE choice. Unbiasedness is a convention, not an optimality property; it is kept because it makes s^2 combine cleanly in the t and F statistics.

5.6.5.3 C. Going deeper

Exercise. C1

Show that a quadratic form $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with $\mathbf{A}$ symmetric is unbiased for σ^2 for every $\boldsymbol{\beta}$ and σ^2 iff $\mathbf{X}^\top \mathbf{A} \mathbf{X} = \mathbf{0}$ and $\text{tr } \mathbf{A} = 1$. Verify that $\mathbf{A} = (\mathbf{I} - \mathbf{H})/(n - p)$ qualifies, and find another $\mathbf{A}$ that does. (Under normality, Theorem 7.4.6 shows that s^2 has the smallest variance among all unbiased estimators.)